



विशाल जाधव सरांचे

VJTech Academy

Inspiring Your Success...

“Multithreading and Exception Handling”

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Live Class Lectures

Program: Create thread by extending Thread class

```
class ThreadX extends Thread
{
    public void run()
    {
        for(int i=1;i<=5;i++)
        {
            System.out.println("From ThreadX:i:"+i);
        }
    }
}
class ThreadY extends Thread
{
    public void run()
    {
        for(int j=1;j<=5;j++)
        {
            System.out.println("From ThreadY:j:"+j);
        }
    }
}
class MainThread
{
    public static void main(String args[])
    {
        ThreadX t1=new ThreadX();
        ThreadY t2=new ThreadY();
        t1.start();
        t2.start();
    }
}
```

Program: Create two threads one will print Even no and another will print odd number between 1 to 10

```
class ThreadX extends Thread
{
    public void run()
    {
        for(int i=1;i<=10;i++)
        {
            if(i%2==0)
            {
                System.out.println("Even Number:"+i);
            }
        }
    }
}
class ThreadY extends Thread
{
    public void run()
    {
        for(int j=1;j<=10;j++)
        {
            if(j%2!=0)
            {
                System.out.println("ODD Number:"+j);
            }
        }
    }
}
class MainThread
{
    public static void main(String args[])
    {
        ThreadX t1=new ThreadX();
        ThreadY t2=new ThreadY();
        t1.start();
        t2.start();
    }
}
```

Program: Create thread by implementing Runnable interface

```
class RunnableX implements Runnable
{
    public void run()
    {
        for(int i=1;i<=5;i++)
        {
            System.out.println("From RunnableX:i:"+i);
        }
    }
}
class RunnableY implements Runnable
{
    public void run()
    {
        for(int j=1;j<=5;j++)
        {
            System.out.println("From RunnableY:j:"+j);
        }
    }
}
class MainThread2
{
    public static void main(String args[])
    {
        RunnableX r1=new RunnableX();
        RunnableY r2=new RunnableY();
        Thread t1=new Thread(r1);
        Thread t2=new Thread(r2);
        t1.start();
        t2.start();
    }
}
```

Program: Set Thread Priority

```
class ThreadX extends Thread
{
    public void run()
    {
        for(int i=1;i<=5;i++)
        {
            System.out.println("From ThreadX:i:"+i);
        }
    }
}
class ThreadY extends Thread
{
    public void run()
    {
        for(int j=1;j<=5;j++)
        {
            System.out.println("From ThreadY:j:"+j);
        }
    }
}
class ThreadZ extends Thread
{
    public void run()
    {
        for(int k=1;k<=5;k++)
        {
            System.out.println("From ThreadZ:k:"+k);
        }
    }
}
class MainThread3
{
    public static void main(String args[])
    {
        ThreadX t1=new ThreadX();
        ThreadY t2=new ThreadY();
        ThreadZ t3=new ThreadZ();
        t1.setPriority(Thread.MIN_PRIORITY);
        t2.setPriority(Thread.NORM_PRIORITY);
        t3.setPriority(Thread.MAX_PRIORITY);
        t1.start(); t2.start(); t3.start();
    }
}
```

Exception Handling

```
class ExceptionDemo
{
    public static void main(String args[])
    {
        int x,y,a=10,b=5,c=5;
        System.out.println("Program Execution Begin...!!!");
        try
        {
            x=a/(b-c);
            System.out.println("Division="+x);
        }
        catch(ArithmeticException e)
        {
            System.out.println("Divide by zero error is occurred");
        }
        System.out.println("Program Execution Ended...!!!");
    }
}
```

Multiple Catch Block

```
class MultipleCatchBlock
{
    public static void main(String args[])
    {
        int x,y,a=10,b=5,c=5;
        System.out.println("Program Execution Begin...!!!");
        try
        {
            x=a/(b-c);
            System.out.println("Division="+x);
        }
        catch(ArithmeticException e)
        {
            System.out.println("Divide by zero error is occurred");
        }
        catch(NullPointerException e)
        {
            System.out.println("NullPointerException is occurred");
        }
        catch(ArrayIndexOutOfBoundsException e)
        {
            System.out.println("ArrayIndexoutOfBoundException is occurred");
        }
        System.out.println("Program Execution Ended...!!!");
    }
}
```

finally keyword

```
class finallyKeywordDemo
{
    public static void main(String args[])
    {
        int x,a=10,b=5,c=5;
        try
        {
            x=a/(b-c);
            System.out.println("Division="+x);
        }
        catch(ArithmeticException e)
        {
            System.out.println("Divide by zero error is occurred");
        }
        finally
        {
            System.out.println("finally block always executed");
        }
    }
}
```


throw keyword

```
class ThrowDemo
{
    void validate(int age)
    {
        if(age<18)
        {
            throw new ArithmeticException("Not eligible for voting");
        }
        else
        {
            System.out.println("You are eligible for voting");
        }
    }
    public static void main(String args[])
    {
        ThrowDemo t1=new ThrowDemo();
        t1.validate(12);
    }
}
```

throws keyword

```
class ThrowsDemo
{
    void myMethod(int num)throws ArithmeticException, NullPointerException
    {
        if(num==1)
            throw new ArithmeticException("ArithmeticException Occurred");
        else
            throw new NullPointerException("NullPointerException Occurred");
    }

    public static void main(String args[])
    {
        try
        {
            ThrowsDemo obj=new ThrowsDemo();
            obj.myMethod(10);
        }
        catch(Exception e)
        {
            System.out.println(e);
        }
    }
}
```

User defined Exception

```
class MyException extends Exception
{
    MyException(String msg)
    {
        super(msg);
    }
}
class UserDefinedException
{
    public static void main(String args[])
    {
        try
        {
            MyException m1=new MyException("User defined exception");
            throw m1;
        }
        catch(MyException e)
        {
            System.out.println(e);
        }
    }
}
```

Write a program to input name and balance of customer and throw a user defined exception if balance less than 1500.

```
import java.util.*;
class MyException extends Exception
{
    MyException(String msg)
    {
        super(msg);
    }
}
class UserDefinedException1
{
    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Customer Name:");
        String name=sc.next();
        System.out.println("Enter Customer Balance:");
        int balance=sc.nextInt();
        if(balance<1500)
        {
            try
            {
                throw new MyException("balance less than 1500");
            }
            catch(MyException e)
            {
                System.out.println(e);
            }
        }
        else
        {
            System.out.println("Customer Name:"+name);
            System.out.println("Customer Balance:"+balance);
        }
    }
}
```

OUTPUT

Enter Customer Name:

Dennis

Enter Customer Balance:

1200

MyException: balance less than 1500

Write a program to input name and salary of employee and throw user defined exception if entered salary is negative.

```
import java.util.*;
class MyException extends Exception
{
    MyException(String msg)
    {
        super(msg);
    }
}
class UserDefinedException2
{
    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Name:");
        String name=sc.next();
        System.out.println("Enter Salary:");
        int salary=sc.nextInt();
        if(salary<0)
        {
            try
            {
                throw new MyException("Salary is Negative");
            }
            catch(MyException e)
            {
                System.out.println(e);
            }
        }
        else
        {
            System.out.println("Name:"+name);
            System.out.println("Salary:"+salary);
        }
    }
}
```

OUTPUT

Enter Name:

Dennis

Enter Salary:

-10000

MyException: Salary is Negative